AI_Driven_Data_Analytics_Internal_Notes:

Module 1 - W1V2 - Introduction to Data Analytics:

What is Data Analytics & it's Process:

- Data Analytics is the ***Process of Examining, Cleaning, Transforming*** and ***Modelling Large Datasets*** to discover meaningful ***Patterns, Insights, Trends*** and support ***Decision-Making***
- **Data Analytics Process:**
    - Raw Data → Data Examination *(Acquisition)* → Data Cleaning & Data Transformation → Data Analysis → Data Visualization → Data Interpretation → Actionable Insights

Data Analytics Life Cycle: (Interview)

The 6-Phase Process

- ***Discovery -*** Understand the Business Problem (Problem Statement) and Define goals
- ***Data Collection -*** Gather data from Databases, API's, Surveys, Sources, Public Records
- ***Data Cleaning & Preparation -*** Clean, Transform & Format data for Analysis
- ***Data Analysis -*** Apply Statistical methods & explore patterns
- ***Data Visualization -*** Create Charts, Dashboards & Visual Stories
- ***Decision & Action -*** Share Insights to your Stakeholders to drive business decisions

Types of Analytics: (Interview)

- ***Descriptive Analytics*** (Tells What Happened)
- ***Diagnostic Analytics*** (Explain Why It Happened)
- ***Predictive Analytics*** (Forecasts What Will Happen)
- ***Prescriptive Analytics*** (Suggests What to Do)

In Data Analytics Perspective - Types of Analytics:

- **Descriptive -** Analyzing Past data to understand Trends
- **Diagnostic -** Identifying reasons for a drop
- **Predictive -** Forecasting next new data based on Trends
- **Prescriptive -** Recommending strategies for Future Development

Applications of Data Analytics:

- Investment Decisions
- Fraud Management
- Risk Assessment
- Customer Behavior
- Demand Prediction
- Pricing Optimization

Real-Time Applications for Data Analytics: (Interview)

- Fraud Detection → Banking
- Sales Analysis → E-Commerce
- Traffic Analysis → Transportation
- Patient Monitoring → Health Care
- Delivery Tracking → Food Delivery
- User Behavior Analysis → social media

Tools for Data Analytics:

- **Excel** - [Data Cleaning, Analysis, Visualizations with built-in Functions and Pivot Tables & Charts]
- **Statistics** - [Science of Collecting, Analyzing, Interpreting and Presenting Data to uncover patterns and Insights]
- **Power BI** - [Business Intelligence Tool for creating Interactive Dashboards and Data Visualizations]
- **MySQL** - [A Relational DataBase Management System used for Storing, Managing and Manipulating Data in an organized and Structured way]
- **Python** - [A versatile Programming Language]
- **Numpy** - [Numerical Python - It is a Library for Numerical Computing and Mathematical Operations]
- **Pandas** - [Panel Data - It is a Data Manipulation Library to form the Data as Series / DataFrame]
- **Matplotlib** - [A Plotting Library used for Creating Static and Interactive Data Visualizations]
- **Seaborn** - [A Statistical Visualization Library used for Informative Plots]
- **Beautiful Soup** - [A Web Scraping library used to extract data from HTML & XML Documents]
- **PySpark** - [Build on Apache Spark, enables scalable MySQL data Processing using JDBC for efficient big data analytics]

Module 1 - W1V3 - Types of Data:

What is Data:

- It is a ***Collection of facts*** such as *Numbers, Words, Quantities, Measurements, Observations or Description of Things*

In Context of Data Science / Data Analysis:

- Data is a ***Collection of Information*** that has been collected for Analysis, which can be used to *Make Decisions, Derive Insights and Build Models*

Types of Data

- **Structure**
    - ***Structured Data -*** Organized in Rows & Columns
        - Excel Sheets
        - SQL Databases
        - Customer Records
        - Financial Reports

      Eg: - Bank Statements

    - ***Semi - Structured Data -*** Has some Structure but doesn't fit neatly in a Table (Contains Tags & Markers)
        - JSON Files
        - XML Data files
        - Emails
        - HTML Pages

      Eg: - WhatsApp Messages, Amazon Products home page

- ***Unstructured Data -*** No Predefined format or model (Majority of World's Data is unstructured)
    - Photos & Videos
    - Audio Files
    - Social Media posts
    - PDF Documents

  Eg: - Customer Reviews in Amazon

- **Observation**
    - ***Cross-Sectional Data -*** Observations of *many subjects* at *one point in Time*
    - ***Time Series -*** Observations of *one subject* across *different Time Periods*
    - ***Panel Data -*** Observations of *many subjects* across *multiple Time Periods*

- **Measurements**
    - ***Numerical / Quantitative Data*** — Numerical Data that can be measured and quantified
        - **Discrete Data** — *Countable values* (Whole Numbers)
        - **Continuous Data** — *Measured values* within the range (Decimal Values)

      Chart Representation based on Types of Data:

        - **Quantitative Data:**
            - Discrete Data — *Pie charts or Bar charts*
            - Continuous Data — *Histogram or Line graphs*

- ***Categorical / Qualitative Data*** — Non - Numerical Data that describes qualities or characteristics
    - **Nominal Data** — Categories without order / No Rank
        - Eg: - Gender, City, Products
    - **Ordinal Data** — Categories with order / Rank

      Eg: -
        - Rating - poor < average < good
        - Education - School < UG < PG
        - Size - Small < Medium < Large
    - **Interval Data —** Ordered with meaningful differences, no true Zero
        - Eg: - Temp, Dates, IQ Scores
    - **Ratio Data —** Ordered with meaningful differences and true Zero
        - Eg: - 0 kg means no weight, height, Duration

- **Availability**
    - ***Primary Data -*** First-Hand Data

      Eg: -
        - Surveys, Interviews, Questionaries, Direct Observations
        - Feedback collecting through Google Forms

    - ***Secondary Data*** - Data Already Collected by someone

      Eg: -
        - Government reports
        - Research papers
        - Customer records
        - Online articles & databases
        - Census population data for Market Analysis